

The logo for SANGUINE is displayed within a blue rounded rectangle. The word "SANGUINE" is written in white, uppercase, sans-serif font. The letter "U" is replaced by a stylized red and white medical test tube with a blue band.

OS13: System Architecture

MITsdm EM.413 - 25 February 2026

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For context:

System Problem Statement:

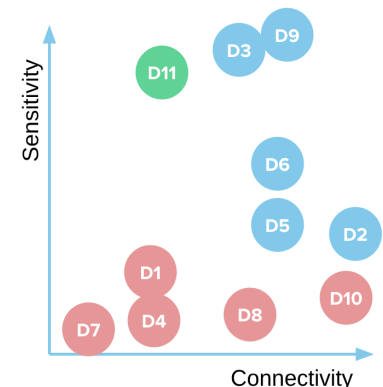
To improve Mayo Clinic lab patient care outcomes at scale
By augmenting physician-facing test results with relevant research
Using an agentic AI clinical librarian

Q1: Architectural Decisions

This table shows our architectural decisions and changes since OS11. We updated almost every decision! Decisions in blue remain, red were closed or removed, and green were added. Options in green are still in play, and options in red are excluded.

ID	Decision name	Scope	Change since OS11	Rationale	Option 1	Option 2	Option 3	Option 4
D1	Compute runtime	Form	Decided	We chose Single-Cloud to align with Mayo Clinic's multi-year strategic partnership with Google Cloud ^[1] .	Edge	On - prem data center	Single - cloud (Google Cloud)	Multi - cloud
D2	Model Origin	Form	Changed options	On GCP, Gemini models are discounted ^[2] . Specialized alternatives may be slightly more performant, but more expensive. We added D11 to choose the model type.	Gemini family only	Other closed source	Train our own	Open source
D3	Research paradigm	Mapping	Changed options	This was the second-most architectural decision in OS11. We added a new option RAG + Fine-Tuning (Piazza @60).	RAG	Fine - tuning	Rely on pre - training	RAG + Fine-Tuning
D4	Communication channels	Mapping	Removed	We discovered in OS11 that D4 is actually neither sensitive nor connected.	EMR only	Web UI only	App only	Web + App
D5	Number of AI roles per test result	Mapping	Changed options	This is architectural, but we reoriented the bucket names and renamed the non-agentic option to be “no AI”, but then removed it because it's outside our SPS.	0 (no AI)	1 (single agent)	Small group (2-5)	Large group (5 - 10)
D6	Data sources	Function	Kept	OS11 showed this is architectural, but we're renaming the options per OS11 feedback to be more strategic.	Mayo internal only	Mayo + single external source	Multiple internal+ external sources	Open Internet only
D7	Coordination structure	Form	Removed	We discovered in OS11 that D7 is actually neither sensitive nor connected.	Agents as Tools	Swarm	Agent Graph / Workflow	No coordination
D8	Clinical system integration	Mapping	Removed	This is not sensitive, and is only strongly connected to one decision we're keeping (D6), meaning this is a design detail.	MCP tools	API tools	N/A	N/A
D9	Clinical validation and release control	Function	Kept	This was the single most architectural decision in OS11. We searched for more options, but these are the main ones.	Rule-based condition	Human - in - the - loop	Tiered release	N/A
D10	Private information handling	Function	Removed	It's best to maximize integration, within compliance constraints. That makes this a constraint, not a decision.	No private exposure	Scoped PHI	No patient - identifiable PHI	N/A
D11	Model Specification	Form	Added	This is a subset of D2, but still architectural because it's sensitive to metrics (M1, M4, M6, and M8) and connected to other decisions (D3, D5, and D10)	Efficient, Text-Only (Flash)	Reasoning, Medium price (2.5 pro)	Multimodal, powerful (3 Pro)	Mix multiple models

These remaining six decisions cover the entire problem space inside our SPS because each concept, as a selection from these options, is architecturally fully specified. As we iterated on this question in parallel with concept generation for Q3, generating concepts suddenly got *much* easier after we got to these decisions. This demonstrated how each possible idea mapped to a selection of these decisions, confirming we had the right set. We also have two decisions each about form, function, and form-function mapping. To compare to OS11, the diagram to the right shows our OS11 Q5 chart (rescaled in response to feedback) color-coded by each decision's status.



Q2: Past Solutions

Epic Systems “MyChart”: This is the incumbent system, with nearly half the current market share [3]. It links test results to patients via Electronic Medical Records (EMR), but population-wide normal ranges are the primary extra context it provides to the physician. Otherwise, the physicians carry the entire cognitive weight of interpreting those results to inform clinical care, and the entire moral responsibility for misinterpretation. Within Epic Systems’ platform, there are nascent AI features, but they’re limited[4]. Several members of our team currently have healthcare that uses Epic MyChart, and can confirm these AI features are not generally available. This differs from our proposed solution through limited clinical diagnostic support to physicians, focusing instead on centralizing and facilitating EMR sharing.

LabCorp Diagnostic Assistant and Insight Analytics: LabCorp's Diagnostic Assistant[5] has a few more AI integrations than Epic’s, but its laboratory diagnostic care is still a rule-based system. On February 23, during this OS’s review, LabCorp announced an expanded collaboration with PathAI[6] an AI based digital pathology platform[7]. Additionally, their Test Finder[8] uses LLMs hosted on Amazon Bedrock to help providers determine which tests to run for their patients. This is the most similar commercial solution employed at scale that we can find related to concepts for Sanguine. There are still distinct differences in execution and while their function has some similarities and the direction of development observed points to solutions we are exploring, our concept’s form of agentic AI to synthesize information for the physician differs from LabCorp’s current system. The recent press releases show the architectural space is quickly evolving, a theme we will return to in Q5.

	Solution Number	Past Solution 1:	Past Solution 2:	Past Solution 3:
ID	Concept name	Epic Systems	Labcorp	Mayo Clinic Current
D2	Model Origin	Not Applicable	Trained internally	Not Applicable
D3	Research paradigm	Not Applicable	RAG Only	Not Applicable
D5	Number of AI roles per test result	0 (no AI)	1 (single agent)	0 (no AI)
D6	Data sources	External sources only	External sources only	Mayo internal only
D9	Clinical validation and release control	Rule - based condition	Rule - based condition	Human - in - the - loop
D11	Model Specification	Not Applicable	Efficient, Text-only (Flash)	Not Applicable

Internal Mayo Clinic Model of Care: The Mayo Clinic Model of Care[9] for internal lab patients uses humans instead of AI-based support. Those experts collaborate and share the cognitive load of patient clinical decisions[10]. This form has many advantages, like enabling the function of peer review in difficult decisions. But, the Mayo Clinic Model of care is expensive, and is not scalable to patients who only come to Mayo Clinic for lab results. Addressing the scale challenge while maintaining the benefits of collaborative peer reviewed care is the crux of this project.

While Mayo Internal is not a third party solution independent to our project or sold to external third parties within healthcare, it is still a cogent solution to review in comparison to our concepts under review. Comparison of a current generation internal project versus the proposed architecture of the next generation of products is a useful architectural practice that can provide significant value to reviewers of a tradespace. Architectural review of past and future systems within a single tradespace along with other external past solutions are a useful audit to ensure justification of the new system is grounded. Our concept is an architectural shift in approach rather than an evolution of these existing paradigms.

Q3: Concepts and Tradespace

We generated these architectures by picking decision options from the “menu” of choices we made in Q1. The fact that this worked so well confirmed we had a good set of architectural decisions.

ID	Name	D2: Model Origin	D3: Research Paradigm	D5: Number of AI roles per test result	D6: Data sources	D9: Release control	D11: Model Specification	Commentary
C1	Maximalist	Gemini family only	RAG + Fine-tuning	Large group (5 - 10)	Multiple internal + external sources	Tiered release	Multimodal, powerful, expensive (3 Pro)	This is the highest-quality, higher-cost solution. These are collectively filtered through a validation and verification stage to ensure compliance and accuracy.
C2	Simplicity	Gemini family only	Rely on pre-training	1 (single agent)	Mayo internal only	Rule-based condition	Efficient, Text-only (Flash)	This concept is a document lookup tool, built using off-the-shelf components.
C3	Max capability within internal-only	Gemini family only	RAG + Fine-tuning	Large group (5 - 10)	Mayo internal only	Human - in - the - loop	Multimodal, powerful, expensive (3 Pro)	Highest precision using a large group of roles with internal-only sources and human validation.
C4	Untrusting	Train our own	Rely on pre-training	Small group (2-5)	Mayo internal only	Tiered release	Multimodal, powerful, expensive (3 Pro)	Internal only trained and deployed agent to collect internal resources. This concept does not trust externally trained models due to the risk of including non-medically relevant training inputs.
C5	Equivalent to patient uploads to ChatGPT	Other closed source	Rely on pre-training	1 (single agent)	Open Internet only	Rule-based condition	Efficient, Text-only (Flash)	Uses readily available models without any data fine-tuning or RAG to give quick responses to data results without context.
C6	Prioritize Customizability	Train our own	RAG + Fine-tuning	Large group (5 - 10)	Multiple internal + external sources	Tiered release	Mix multiple models	This concept makes all the most customizable architectural decisions. If we wanted to strategically expand this into a product platform, we would value this extensibility.
C7	Single Expert	Other closed source	RAG + Fine-tuning	1 (single agent)	Mayo internal only	Human - in - the - loop	Multimodal, powerful, expensive (3 Pro)	This concept centers on a single powerful agent. By maximizing performance and using human-in-the-loop validation, it creates a narrow but deep expert that humans can consult.
C8	Low-cost Efficient	Other closed source	RAG + Fine-tuning	1 (single agent)	Mayo internal only	Rule-based condition	Mix multiple models	This concept leverages customization to drive cost efficiency by selecting the right models for each task.
C9	Embedded Knowledge	Train our own	Rely on pre-training	1 (single agent)	Multiple internal + external sources	Tiered release	Multimodal, powerful, expensive (3 Pro)	Uses its own multimodal model that embeds all internal sources, instead relying on RAG to manage interpretation sources.
C10	Compliance Safe	Train our own	RAG Only	Large group (5 - 10)	Mayo internal only	Human - in - the - loop	Multimodal, powerful, expensive (3 Pro)	Built on a proprietary multimodal model, using curated internal sources through RAG, coordinated across multiple specialized roles, and reviewed by human validation to ensure a compliance-first approach.

ID	Name	Cost Score	Accuracy Score
C1	Maximalist	40	49
C2	Simplicity	15	29
C3	Max capability within internal-only	38	48
C4	Untrusting	34	41
C5	Equivalent to patient uploads to ChatGPT	19	25
C6	Prioritize Customizability	43	50
C7	Single Expert	35	43
C8	Low-cost efficient	19	33
C9	Embedded Knowledge	36	41
C10	Compliance Safe	41	48

Metrics

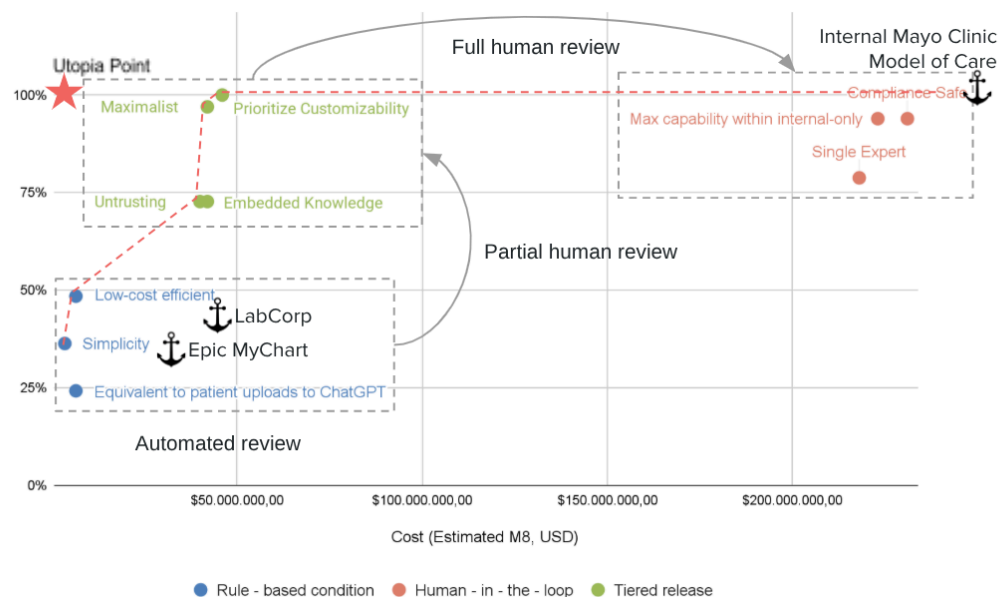
We choose one performance metric and one cost metric. For performance, we choose M6: clinical accuracy as measured by the agreement rate between our AI output and a human expert. This is the most important metric, as shown in OS11 Q3, because incorrect interpretation can harm patients and create legal and regulatory risk. For cost, we choose M8: cost per interpreted result. This cost metric captures the cost/benefit of the system, and is feasible because cloud compute prices are generally pay-what-you-use. This table shows each historical and new concept along each metric.

Model

We made a high-level model by estimating the impact of each option on cost (in dollars) and on accuracy (in arbitrary units from 0 to 10) using arbitrary units. Then, we define a concept's score to be the sum of the costs and accuracy amounts of each of that concept's decisions. For example, for D5 (agent group size), using non-agentic AI (option 1) adds 1 point to accuracy and \$0 to cost, while choosing a large group of agents (option 4) adds 9 points to accuracy and \$7M to cost. Full scores, including rationale for each number, are in Appendix 2. We choose this modeling technique because it's easy to tune and evaluate. The scores for each option, and the rationale and citations for each score, are in an appendix. We prioritized making a lightweight model, so it **intentionally does not account for interaction between architectural decisions**. We expect a future OS to involve evaluating concepts at that level of detail.

Tradespace and Insights

The tradespace highlights the tension between the cost per interpreted clinical outcome and the clinical accuracy obtained. The clusters (from bottom left to top right) were organized so that architectural decisions structurally constrain system performance along the Pareto frontier. To anchor our tradespace in past designs, we show the Q2 concepts with an anchor icon (⚓). **The first region** is defined by **low cost and low accuracy**, aligning with Epic Systems, whose deterministic rules limit accuracy. These concepts bring together less advanced models, minimal validation, little role decomposition, and restricted use of clinical sources. Performance is limited by insufficient clinical knowledge, with decisions on “research paradigm” and “data sources” dominating and defining the baseline architecture. **The intermediate region** corresponds to **moderate cost and higher accuracy**, aligning with Labcorp by showing increased accuracy with model assistance. These concepts use more consistent retrieval information, moderate role decomposition, internal clinical sources, and tiered validation. The gain in accuracy is due to role decomposition in clinical interpretation. The dominant decision is “number of AI roles per result”. This region defines the efficient frontier, as accuracy improvements are achieved without intensive human intervention. **The third region** is associated with **clinical assurance** and aligns with the current Mayo Clinic solution, which involves validating physicians' clinical interpretations, resulting in high reliability but with high costs. These concepts have more complex or proprietary models, multiple coordinated roles, a greater number of clinical sources, and more rigorous review of results, with more frequent human-in-the-loop intervention. The dominant decision is “clinical validation and release control”, as the clinical reasoning decisions already employ the highest clinical accuracy scores. The increase in costs is more closely linked to the validation and release of clinical interpretation than to improvements in the clinical reasoning process.



Q4: Decompositions

This decomposition describes the system's **functions** for delivering safe, understandable lab-result explanations and appropriate escalation for concept 7. Functions are decomposed from **Level 0** → **Level 1** → **Level 2** using a “two down” approach. L0 is : Provide safe, evidence-based, patient-appropriate interpretation of lab results with appropriate escalation and follow-up. We **prefer the second L1 decomposition** because it more insightfully minimizes connections across components.

L1A (1 Down) L1B (after 2 Downs , 1 UP)

Function	1	2	3	4	5	6	7
1. Manage access, consent, and privacy							
2. Manage user interaction and intent	X						
3. Acquire patient clinical context	X	X					
4. Interpret lab results and clinical risk			X				
5. Retrieve and manage supporting evidence				X			
6. Compose patient-facing explanation and guidance				X	X		
7. Govern release, escalation, and follow-up	X						

Function	1	2	3	4	5
1 – Authenticate & Authorize Access					
2 – Capture Intent & Retrieve Lab Data	X				
3 – Contextualize & Interpret Results	X	X			
4 – Assemble Evidence-Based Explanation			X		
5 – Govern Release & Escalation	X			X	

L2 (2 downs) + 1Up (colored regions) | Note: X in row *i*, col *j* means "row *i* depends on col *j*."

Function	1.1	1.2	1.3	1.4	1.5	2.1	2.2	2.3	2.4	2.5	3.1	3.2	3.3	3.4	3.5	3.6	4.1	4.2	4.3	4.4	4.5	4.6	5.1	5.2	5.3	5.4	5.5	6.1	6.2	6.3	6.4	6.5	6.6	6.7	7.1	7.2	7.3	7.4	7.5	7.6	7.7	7.8	7.9	
1.1 Authenticate user identity																																												
1.2 Determine user role & relationship	X																																											
1.3 Verify consent & permitted scope	X	X																																										
1.4 Authorize access to data elements	X	X	X																																									
1.5 Enforce privacy constraints on outputs	X	X	X	X																																								
2.1 Capture user inquiry	X																																											
2.2 Detect user intent category						X																																						
2.3 Extract key entities						X	X																																					
2.4 Detect ambiguity/missing details						X	X	X																																				
2.5 Request clarifying info						X	X	X	X								X																											
3.1 Retrieve relevant lab results			X	X					X	X	X																																	
3.2 Retrieve reference ranges/units/flags			X	X					X	X																																		
3.3 Retrieve prior related labs (trends)			X	X					X	X																																		
3.4 Retrieve relevant patient factors			X	X					X	X																																		
3.5 Validate/normalize data											X	X	X	X																														
3.6 Identify context limitations											X	X	X	X	X																													
4.1 Compare to reference ranges & identify abnormalities											X	X	X																															
4.2 Determine severity tier											X	X	X																															
4.3 Perform trend assessment											X	X	X	X						X																								
4.4 Identify benign explanations/confounders											X	X	X																															
4.5 Identify red-flag patterns											X	X	X	X	X	X																												
4.6 Identify when interpretation is unsafe (insufficient context)											X	X	X	X	X	X	X	X	X	X																								
5.1 Formulate evidence needs																			X	X	X	X			X																			
5.2 Retrieve clinical references																																												
5.3 Rank/filter evidence																																												
5.4 Extract evidence statements for lay explanation																																												
5.5 Track evidence provenance metadata																																												
6.1 Select explanation structure																																												
6.2 Translate interpretation into plain language																																												
6.3 Explain meaning/causes (non-diagnostic)																																												
6.4 Provide safe next-step guidance																																												
6.5 Provide urgency guidance																																												
6.6 Add citations/attribution																																												
6.7 Tailor to user preferences																																												
7.1 Screen draft for unsafe/disallowed advice																																												
7.2 Check compliance with scope/consent/privacy	X	X	X	X	X																																							
7.3 Assess uncertainty & need escalation																																												
7.4 Decide release path																																												
7.5 Prepare escalation package																																												
7.6 Route case for human review																																												
7.7 Incorporate review outcome											</																																	

The guiding principle of DSM-based modularization is to cluster functions that share the highest density of dependencies into the same module, thereby minimizing off-diagonal interactions and producing tight diagonal banding. L1B adheres to this principle more faithfully than L1A. In the L2 DSM, evidence retrieval sub-functions (5.1–5.5) and explanation composition functions (6.1–6.7) exhibit strong mutual dependencies — evidence provenance feeds directly into citation placement, while explanation structure selection drives which evidence is retrieved and filtered. Splitting these across two L1 blocks, as L1A does, forces high-traffic information flow across a module boundary, violating the very coupling minimization that modularization seeks to achieve. Similarly, "Retrieve relevant lab results" (3.1) shows stronger dependency marks with intent extraction (2.3) and entity capture (2.2) than with the remaining data retrieval functions in Block 3, indicating that the natural dependency cluster crosses L1A's block boundary. L1B resolves both of these by following the architecture — not the process narrative. The result is a five-block decomposition with fewer inter-block interfaces, tighter internal cohesion within each module, and a cleaner lower-triangular structure in the L1 DSM, all of which reduce integration complexity and better satisfy the modularity condition that each block should be maximally independent from every other while remaining internally coupled.

Q5: Platform Strategy

No, we explicitly forego a platform strategy for Sanguine. If our initial concept architecture is successful, leading to a widespread distillation of the role of a clinical librarian across Mayo Clinic and external physicians at large, it could warrant future development into a platform. In that scenario, additional effort to establish how interactions across boundaries are maintained and the infrastructure to support would be required. But, that is premature at this stage. Sanguine is a concept that seeks to usurp current legacy methods that are technologically stagnant by employing a rapidly developing new technology through the use of agentic AI models to enable scale. The turbulent nature of agentic AI creates questions regarding the durability of interfaces produced to encompass them and whether their cost is warranted. At this stage it is clear that our product strategy would be better served by forgoing “platformization” and the investments required to realize that reality.

The current phase within the Technology S-Curve leads to state instability. The rapid changes within multiple different industries reckoning with the deployment of AI illustrates turbulent instability^{[11][12][13]} as adoption goes through the steepest slope of the technology S-Curve^[14]. While our architectural decisions for Sanguine are solidifying, the long term evolutionary form of this architecture is more amorphous. Due to scope and diminishing returns in impact during an initial product architecture, we have already excluded specific market variants including: patient facing agentic AI chat interfaces, agentic AI data auditor who facilitates data summation from disparate sources, patient action planning roles, and explicit collaborative links to Mayo Clinic’s existing platform^[15]. These are all natural areas this system architecture could expand to become part of a platform strategy in a future iteration, though at this time it would be premature and increase cost. Implementation and time to market are key factors that demand a constrained system boundary; any expansion would be detrimental to Sanguine’s ability to prove value during this initial phase.

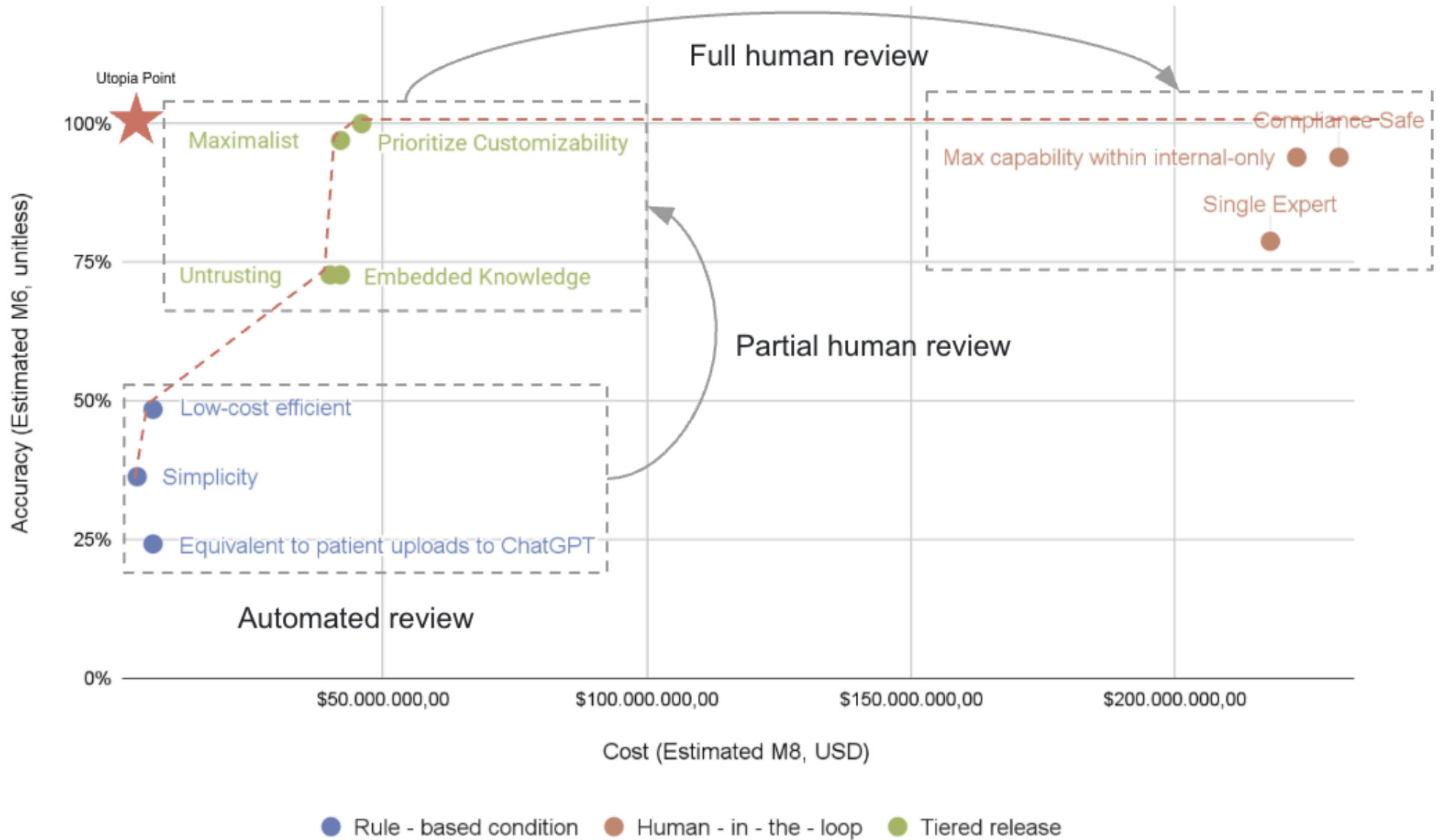
System architecture evolution is likely to occur through generational reuse. Hardening interfaces through platform development is more likely to add brittle fragility rather than lasting scaffolding. Instead of investing into a product platform without a strong understanding of whether a majority of the system components will be reusable by other systems, we will launch an MVP. Later, we could transition this into a “beachhead strategy” (*as discussed in lecture SA13 slide 26*) with Sanguine that can then be used to determine next actions once stable. This gives us flexibility to allow future systems to eventually form into a product platform if reasonable, but the current state lacks the architectural solidity to justify investment into a platform. Likely the system architecture will have a series of generational reuses until iteration leads to calcification of understanding and clear stability in interfaces. We choose to instead focus our product market on physician support, which addresses the market need in a more constrained manner but exposes less risk, including liability risks. A product platform could eventually form from the evolution of this system, but trying to attend to a larger product market of use cases beyond the role of a clinical librarian is premature. The current state lacks the architectural solidity to justify investment into a platform until a series of generational reuses or iteration leads to calcification of understanding and clear stability in interfaces.

Constraints that currently exist may cease to exist during the system development. Some of our key decisions rely on current understandings of performance model performance. This is a rapidly changing field. A recently published study in Nature titled *Toward expert-level medical question answering with large language models (Jan 2026)* said, “The LLM landscape is rapidly evolving, necessitating careful interpretation of our findings within this dynamic context. Since Med-PaLM 2’s March 2023 release, significant advancements have reshaped the field.”^[16] There is no promise that advancements similar to those observed by the transition after transformer architectures were introduced could not occur again during the development of this system. Further new constraints tied to introduction of regulation related to the deployment of AI within the medical industry could create constraints further complicating the development of this system. We need flexibility to accommodate changes during this period of evolution, which could necessitate some willingness to abandon key decisions within the system. While we have some grounding in our decisions and strategy at this point in time, we see potential for the decisions: Model Origin (D2), Research paradigm (D3), Data Sources (D6), and Clinical validation and release control (D9), being sensitive to this turbulent period of advancement. This could reshape and redefine what concepts are best aligned with the Mayo Clinic Model of Care.

Q6: Schedule Briefing

Our final briefing is scheduled for Thursday May 7 at 11AM Eastern. Our project sponsor Chris Garcia will be there, and we are arranging for other attendees to join too.

Appendix 1: Larger Tradespace



Appendix 2: Model Scores and Costs

This table shows the estimated cost and accuracy score of each option, and our rationale for each.

ID	Decision name	Option	Annual cost	Cost rationale	Accuracy score (0-10, 10 is best)	Accuracy rationale
D2	Model Origin	Gemini family only	\$1.000.000,00	Bundle pricing and integration reduce data transfer overhead.	7	Generalist with no clinical specialization.
D2	Model Origin	Other closed source	\$2.000.000,00	Separated providers increase licensing and data transfer overhead.	6	Similar performance with no clinical specialization.
D2	Model Origin	Trained internally	\$12.000.000,00	Continuous training, model development, and infrastructure operations impact cost highly.	8	Can be specialized to interpret patterns and leverage existing data sources.
D2	Model Origin	Open source	\$4.000.000,00	With no license, or hosting or managed services will be required, adding more cost to them	5	Weaker clinical reasoning without tuning due to narrow parameter's scope.
D2	Model Origin	No model used	\$6.000.000,00	Requires a deterministic rule engine and manual knowledge operation.	3	Limited to predefined rules, poor interpretation on ambiguous cases.
D3	Research paradigm	RAG Only	\$2.000.000,00	Retrieval pipeline, embeddings and vector storage add operations and hosting costs.	7	Anchors the interpretation to clinical references, reducing hallucination.
D3	Research paradigm	Fine - tuning Only	\$3.000.000,00	Dataset curation and retraining cycles increase operational costs.	7	Improves model consistency but lacks external clinical references.
D3	Research paradigm	RAG + Fine - tuning	\$5.000.000,00	Combines retraining with retrieval infrastructure.	8	Combines evidence retrieval with stable interpretation patterns.
D3	Research paradigm	Rely on pre - training	\$0,00	No retrieval or retraining costs required.	3	Only uses generic knowledge from models, with unreliable clinical background.
D3	Research paradigm	No model used	\$4.000.000,00	Manual knowledge base maintenance required.	3	Can't interpret complex clinical cases.

ID	Decision name	Option	Annual cost	Cost rationale	Accuracy score (0-10, 10 is best)	Accuracy rationale
D5	AI roles per test result	0 (non - agentic AI)	\$0,00	Single inference with no coordination.	1	No cross-checking of interpretation.
D5	AI roles per test result	1 (single agent)	\$1.000.000,00	One inference chain per result.	5	Limited validation comparison.
D5	AI roles per test result	Small group (2-5)	\$3.000.000,00	Multiple inference calls and coordination between agents.	8	Independent roles can verify interpretations.
D5	AI roles per test result	Large group (5 - 10)	\$7.000.000,00	High token usage and coordination latency.	9	Additional agents can produce similar reasoning, adding incremental accuracy gain.
D6	Data sources	Mayo internal only	\$0,00	Existing institutional data curated.	6	Interpretation limited to local references.
D6	Data sources	Mayo + single external source	\$1.500.000,00	External access processing overhead.	8	Adds independent reference improving accuracy and reliability.
D6	Data sources	Multiple internal + external sources	\$4.000.000,00	Multiple retrieval pipelines and standardization required.	9	Enables cross-evidence comparison and adds differential analysis.
D6	Data sources	Open Internet only	\$2.000.000,00	Filtering and ranking required.	3	High noise and non-curated medical references.
D6	Data sources	External sources only	\$3.000.000,00	Integration and possibly licensing costs to the sources with no internal data usage.	2	Lacks patient-specific context.
D9	Clinical validation and release control	Rule - based condition	\$500.000,00	Automated checks with minimal operational effort.	5	Detect more obvious errors.
D9	Clinical validation and release control	Human - in - the - loop	\$200.000.000,00	Requires physician review time per interpreted result.	10	Expert review with corrections before release.
D9	Clinical validation and release control	Tiered release	\$15.000.000,00	Partial review and escalation workflow.	8	High-risk cases are validated only.
D11	Model Specification	Efficient, Text-only (Flash)	\$1.000.000,00	Lowest latency and lowest token cost for inference.	3	Limited reasoning depth.
D11	Model Specification	Reasoning, Medium price (2.5 pro)	\$4.000.000,00	Higher inference cost and latency.	7	Support structured clinical reasoning.
D11	Model Specification	Multimodal, powerful, expensive (3 Pro)	\$10.000.000,00	Highest context processing cost.	8	Improves evidence integration.

ID	Decision name	Option	Annual cost	Cost rationale	Accuracy score (0-10, 10 is best)	Accuracy rationale
D11	Model Specification	Mix multiple models	\$3.000.000,00	Routing overhead to avoid using expensive models for all situations.	8	Only complex cases are handled to advanced models.
D11	Model Specification	No model used	\$5.000.000,00	Manual workflows and rule maintenance.	1	No interpretation capability.

Cost estimates were based on an annual volume of 26 million laboratory tests, as reported by Mayo Clinic Laboratories ^[17]. The values presented represent the annual increase in the total system cost, estimated in order of magnitude. The cost structure considered includes:

- Dedicated technical team (ML engineering, MLOps, DevSecOps, SRE) ^[18]
- Clinical review time, when applicable ^{[19], [20]}
- Cloud infrastructure and computing capacity ^{[21], [22]}
- Model inference costs and API consumption ^{[23], [24]}
- Data integration, governance, compliance, and lifecycle management

Appendix 3: AI Disclosure

Parts of the cost modeling, reasoning, and refinement in this document were developed with the aid of a large language model (LLM).

All numerical estimates, modeling assumptions, and architectural interpretations were reviewed and adjusted by the authors.

No proprietary or confidential institutional information was used during the review of these items. Google's Gemini was used to review drafts of this assignment and act as the role of a MITsdm TA providing feedback on sections with strong arguments and areas requiring improvement. All work in response to this feedback is the author's own.

Appendix 4: Works Cited

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